

Deuterium and carbon-13 kinetic isotope effects for the reaction of OH with CH₄

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Interpolated variational transition state theory calculations with centrifugal-dominant, small-curvature tunneling coefficients have been carried out for the case of the deuterium kinetic isotope effect (KIE) in the reaction $\text{OH} + {}^{12}\text{CD}_4 \rightarrow \text{HDO} + {}^{12}\text{CD}_3$ and for the ${}^{13}\text{C}$ KIE for the reaction $\text{OH} + {}^{13}\text{CH}_4 \rightarrow \text{H}_2\text{O} + {}^{13}\text{CH}_3$. The interpolated variationally optimized generalized transition states predict notably different nontunneling KIEs than the conventional ones, and factorization analyses of the KIEs are presented to illustrate the origin of the differences. The zero-point energies at the variational transition states differ from those at the saddle point by up to 0.19 kcal/mol for the $\text{OH} + {}^{12}\text{CD}_4$ reaction and by up to 0.34 kcal/mol for the $\text{OH} + {}^{13}\text{CH}_4$ reaction. The incorporation of multidimensional tunneling effects partly cancels the effect of variational optimization of the transition state.

I. INTRODUCTION

Kinetic isotope effects (KIEs) provide unique insight into the dynamics of chemical reactions.¹⁻⁸ The present paper is concerned with the theoretical interpretation of KIEs on the rate constants of OH radical reactions with CH₄ in the gas phase and with comparison of calculations to experiment⁹⁻¹² for these KIEs. The deuterium KIE for the reaction of OH with CH₄ is defined in this article as the ratio $k_{\text{H}}/k_{\text{D}}$, where k_{H} is the rate constant for the unsubstituted methane and k_{D} is the rate constant for the one in which all four protons are substituted by deuterium. The ${}^{13}\text{C}$ KIE is denoted k_{12}/k_{13} , where k_{12} is the rate constant for the normal methane and k_{13} is the rate constant for the one in which the ${}^{12}\text{C}$ atom is substituted by ${}^{13}\text{C}$. (By definition, a KIE is "normal" if it is greater than unity when the rate constant for the lighter isotope is in the numerator, and it is an "inverse" isotope effect if it is less than unity.) The theoretical treatment used in the present study is the recently developed method of interpolated variational transition state theory^{13,14} (IVTST) with tunneling included by a recently developed multidimensional semiclassical approximation called the centrifugal-dominant small-curvature semiclassical adiabatic ground-state approximation,^{15,16} or—for brevity—the small-curvature tunneling (SCT) approximation.

Conventional transition state theory (TST) is the most widely applied tool for the interpretation of KIEs.¹⁻⁵ However, in many cases rate constant results obtained using variationally optimized generalized transition states and including multidimensional tunneling effects are considerably more accurate.¹⁷⁻²⁸ In this article, we use VTST to calculate deuterium and ${}^{13}\text{C}$ KIEs for the gas-phase reaction of OH with CH₄.

The most recent advance in the practical applicability of VTST is the development of IVTST, in which high-quality calculations of the potential energy, its gradient, and its Hessian are performed at three to five selected points along the reaction path, and all information needed

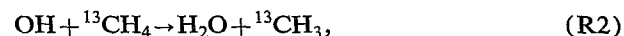
for VTST calculations is interpolated from this data without using a global potential energy function.^{13,14} Here, we use the highest-order version of the IVTST hierarchy,¹³ namely, a second-order global version of interpolated canonical variational transition state theory, to locate the transition state dividing surface. In particular, we base our results on accurate *ab initio* calculations (some from Ref. 14 and some reported here) for the structures, energetics, and vibrational frequencies of the reactants, products, and saddle point of the reactions studied, and we also use results at the same level for two extra points along the reaction path—one on each side of the saddle point. Moreover, we include multidimensional tunneling effects in the transmission coefficient based on the new SCT^{15,16} method.

The energy and force constants of the optimized variational transition state for the unsubstituted reaction may be significantly different from those for the isotopically substituted reaction, whereas in conventional TST, these properties are not influenced by isotopic substitution. Moreover, the effective potential used for the calculation of multidimensional tunneling effects includes all of the vibrational frequencies, even those associated with spectator modes.^{6-8,15-17,21,27,29-31}

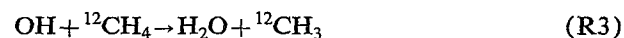
The reactions whose rate constants are calculated here are



and



and results from the previous study¹⁴ of the reaction



are used to compute the KIEs.

The deuterium KIE for reaction (R1) has previously been studied experimentally by Gordon and Mulac,⁹ whereas the ${}^{13}\text{C}$ KIE for reaction (R2) has been examined in detail by Rust and Stevens,¹⁰ Davidson *et al.*,¹¹ and Cantrell *et al.*¹² Reaction (R2) is of special interest, since

in order to understand the sources and sinks of atmospheric methane,³² the methane budget must be known; and one way to study the methane budget is through the use of stable isotopes of carbon as proposed by Stevens and Rust.³³

For reaction (R3), the rate constants have been measured by using pulsed photolysis-laser induced fluorescence³⁴ (PP-LIF) and other methods,³⁵ and reasonable agreement (within a factor of 2) with experiment was obtained in previous¹⁴ IVTST calculations.

The level of electronic structure we use is second-order Møller-Plesset-Scaling all correlation-energy (MP-SAC2). Our MP-SAC2//MP2/adjusted correlation-consistent polarized-valence triple-zeta (adj-cc-pVTZ)³⁶⁻³⁸ electronic structure calculations¹⁴ predict a classical exoergicity³⁹ of 13.27 kcal/mol (compared to the experimental value^{14,40} of 13.27 ± 0.29 kcal/mol) and classical barrier heights³⁹ of 7.37 and 20.64 kcal/mol for the forward and reverse (R1), (R2), and (R3) reactions.

In Sec. II, we briefly review the scaling all correlation-energy (SAC) method for calculating the total absolute energies of the chemical systems under study and the conventional TST theory for computing the rate constants. Then we contrast TST to VTST theory.

II. METHODS

II.A. Electronic structure calculations

The motivation of the SAC³⁷ method for large basis sets, such as those used for the calculations here, is to make the calculated exoergicity of the reaction under study agree perfectly with experiment so that the treatment of reactants and products is balanced, and a set of forward and reverse rate constants consistent with the equilibrium constant can be obtained. The calculated scaling factor $\mathcal{F}_2$ for the dissociation energies of both the O-H and C-H bonds in our system, and for the adj-cc-pVTZ basis set, is equal to 0.967.¹⁴ Further details of the SAC method are given elsewhere.³⁷

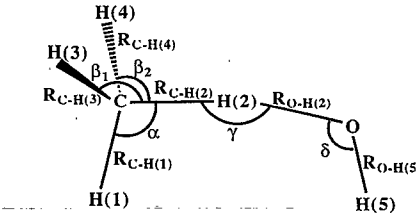
The starting point for our large-scale calculations is the correlation-consistent polarized-valence triple-zeta (cc-pVTZ) basis set of Dunning.³⁸ For C and O, this is a (10s5p2d1f) primitive Gaussian basis set contracted to [4s3p2d1f], whereas for H it is a (5s2p1d) primitive set contracted to [3s2p1d]. Pure *d* and *f* functions are used in the *d* and *f* shells, i.e., the *d* functions have five components and the *f* functions have seven. Here, as before,¹⁴ we used an adjusted version of this basis set where the same scaling factor applies to the computed dissociation energies of both the O-H and C-H bonds. This adjusted basis, called adjusted cc-pVTZ (adj-cc-pVTZ), is explained in detail elsewhere;¹⁴ it treats the electron correlation in both the forming and the breaking bonds of the above systems on an equal footing in order to obtain an accurate barrier height for both directions of each reaction.

A mass-scaled coordinate system^{31,41} was defined as

$$x_{i\gamma} = (m_i/\mu)^{1/2} R_{i\gamma}, \quad (1)$$

where m_i is the mass of atom *i*, μ is the reduced mass of the relative translational motion of reactants, which equals

TABLE I. The geometry of the nonstationary points on the OH + ¹²CD₄ and OH + ¹³CH₄ reaction paths.^a



<i>s</i> (<i>a</i> ₀)	OH + ¹² CD ₄		OH + ¹³ CH ₄	
	-0.005	+0.005	-0.005	+0.005
<i>R</i> _{C-H(1)}	1.0804	1.0801	1.0804	1.0801
<i>R</i> _{C-H(2)}	1.1853	1.1955	1.1834	1.1974
<i>R</i> _{C-H(3)}	1.0808	1.0798	1.0810	1.0796
<i>R</i> _{C-H(4)}	1.0803	1.0803	1.0804	1.0802
<i>R</i> _{O-H(2)}	1.3093	1.2978	1.3111	1.2959
<i>R</i> _{O-H(5)}	0.9697	0.9698	0.9697	0.9698
α	108.20	108.06	108.22	108.03
β_1	105.47	105.50	105.47	105.50
β_2	105.55	105.42	105.57	105.40
γ	169.56	169.61	169.55	169.62
δ	96.99	97.13	97.00	97.12
ϕ_1^b	120.20	120.26	120.20	120.27
ϕ_2^c	120.29	120.18	120.31	120.16

^aBond lengths in Ångstroms and, bond and dihedral angles in degrees.

^b ϕ_1 is the dihedral angle between the H(1)-C-H(2) and H(3)-C-H(2) planes.

^c ϕ_2 is the dihedral angle between the H(1)-C-H(2) and H(4)-C-H(2) planes.

9.202 amu for reaction (R1), 8.509 amu for reaction (R2), and 8.251 amu for reaction (R3), and *R*_{*i*γ} is a Cartesian coordinate for $\gamma = x, y, \text{ or } z$. The geometries of the nonstationary points along the minimum energy path^{31,42} (MEP) on either side of the saddle point were determined by taking steps of length $\pm 0.005a_0$ in the direction of the mass-scaled eigenvector corresponding to the saddle-point imaginary frequency for each reaction. (A negative sign denotes a step towards reactants and a positive sign denotes a step towards products.)

The geometries of all stationary points, i.e., minimum energy structures and the saddle point, were fully optimized¹⁴ at the MP2 level of theory with the adj-cc-pVTZ basis set. *C*₁ point group symmetry was used for each species as the initial geometry of the optimization procedure and all the *Z*-matrix⁴³ parameters were fully optimized. The resulting point group symmetries are *C*_{∞v} for OH, *D*_{3h} for CH₃, *T*_d for CH₄, *C*_{2v} for H₂O, and *C*_s for the CH₃-H-OH saddle point. The extra points along the reaction path on each side of the saddle point have a *C*₁ symmetry. The absolute energy and geometry of the stationary points, which are the same for all three reactions (R1)-(R3), and of the nonstationary points for reaction (R3) are given in a previous paper by the present authors,¹⁴ whereas the geometry of the nonstationary points for reactions (R1) and (R2) are listed in Table I. As mentioned above, steps of 0.005*a*₀ length were used for calculating the nonstationary points geometry for reactions (R1) and (R2), since that value gave stable results in Ref. 14 and,

because of our choice of the scaling mass μ , using it here gives similar values for breaking ($R_{C-H(2)}$) and forming ($R_{O-H(2)}$) bond lengths at the nonstationary points as were obtained¹⁴ for breaking ($R_{C-H(2)}$) and forming ($R_{O-H(2)}$) bond lengths of reaction (R3), respectively, on each side of the saddle point. The MP-SAC2//MP2/adj-cc-pVTZ energy of the backward point ($s = -0.005a_0$) for reactions

(R1)–(R3), was higher than the saddle point ($s=0$) MP-SAC2//MP2/adj-cc-pVTZ energy because the saddle point was not optimized at the SAC level, and therefore the original SAC procedure is not useful along the reaction path. Consequently, at the two nonstationary points for each reaction, the scaling was accomplished in the following way:

$$E(\text{scaled-MP2/adj-cc-pVTZ}, s) = E(\text{MP2/adj-cc-pVTZ}, s) + [E(\text{MP-SAC2//MP2/adj-cc-pVTZ}, s=0) - E(\text{MP2/adj-cc-pVTZ}, s=0)]. \quad (2)$$

[See the supplementary material⁴⁴ for absolute energies of the nonstationary points for reactions (R1) and (R2).]

All points for reactions (R1) and (R2) were characterized by vibrational analysis, and the vibrational frequencies for the equilibrium species, i.e., reactants and products calculated at the MP2/adj-cc-pVTZ level are in Table II. Table II also lists experimental^{40,45–48} values. It should be

noted that the calculated frequencies are harmonic values and the differences between experimental harmonic and fundamental frequencies may be as large as the differences between theory and experiment in the table. The transition state frequencies at the nonstationary points for reactions (R1) and (R2) calculated at the MP2/adj-cc-pVTZ level are given in Table III. Table IV lists the vibrational frequencies at the MP2/adj-cc-pVTZ level for reactions (R1) and (R2), respectively. For interpretative reasons, we also include in Table IV the vibrational frequencies at the saddle point ($s=0$) and the interpolated ones at the generalized transition states for three temperatures, namely, 300, 600, and 1500 K, and at $s=0.1$ and $0.8a_0$. The vibrational frequencies of reaction (R3) are given elsewhere.¹⁴

Since unrestricted Hartree–Fock (UHF) wave functions are not spin eigenfunctions, we monitored the expectation value of S^2 in all UHF³⁶ and MP2 wave functions (where S denotes electron spin angular momentum in units of $\hbar$). For doublets, S^2 was always in the range 0.750–0.781, where 0.750 is the correct value. Therefore, spin contamination is not severe. We note for completeness

TABLE II. Vibrational frequencies (cm^{-1}) of the equilibrium structures for the reactions $\text{OH} + {}^{12}\text{CD}_4 \rightarrow \text{HDO} + {}^{12}\text{CD}_3$ and $\text{OH} + {}^{13}\text{CH}_4 \rightarrow \text{H}_2\text{O} + {}^{13}\text{CH}_3$.

	Mode	MP2/adj-co-pVTZ	Harmonic (Expt.)	Fundamental (Expt.)
OH	σ_s	3759 (5.37) ^a	3735 ^b	3570 ^b
H ₂ O	a_1	3795	3834 ^c	3651 ^b
	a_1	1663	1648 ^c	1595 ^b
	b_2	3923 (13.41) ^a	3935 ^c	3756 ^b
HDO	a_1	2802	2824 ^d	2724 ^b
	a_1	1458	1440 ^d	1403 ^b
	b_2	3861 (11.61) ^a	3890 ^d	3707 ^b
¹² CD ₃	a'_1	2264	n/a ^e	2136 ^f
	a'_2	389	n/a	458 ^f
	e'	2513	n/a	2381 ^f
	e'	1067 (14.03) ^a	n/a	1026 ^f
¹³ CH ₃	a'_1	3201	n/a	3006 ^f
	a'_2	498	n/a	604 ^f
	e'	3358	n/a	3150 ^f
	e'	1445 (19.02) ^a	n/a	1390 ^f
¹² CD ₄	a_1	2188	2140 ^g	2118 ^g
	e	1132	1120 ^g	1092 ^g
	t_2	2381	2336 ^g	2260 ^g
	t_2	1027 (20.98) ^a	1034 ^g	998 ^g
¹³ CH ₄	a_1	3093	3026 ^g	2919 ^g
	e	1600	1583 ^g	1533 ^g
	t_2	3204	3146 ^g	3010 ^g
	t_2	1350 (28.53) ^a	1359 ^g	1303 ^g

^aValues in parentheses are total harmonic zero-point energies in kcal/mol.

^bReference 40.

^cReference 45.

^dReference 46.

^en/a—not available.

^fReference 47.

^gReference 48.

TABLE III. Saddle point harmonic vibrational frequencies (cm^{-1}) at the MP2/adj-co-pVTZ level for reactions (R1) and (R2), and the respective harmonic zero-point energies (kcal/mol).

Reaction	OH + ¹² CD ₄	OH + ¹³ CH ₄
a'	3748	3748
	2415	3243
	2234	3128
	1123	1520
	1071	1408
	979	1293
	869	930
	694	756
a''	230	356
	2418	3247
	1077	1492
	892	1201
	276	316
a'	38	40
	1349 <i>i</i>	1830 <i>i</i>
ZPE	25.82	32.42

TABLE IV. Transition state and generalized transition states vibrational frequencies (cm^{-1}) along the $\text{OH} + {}^{12}\text{CD}_4$ and $\text{OH} + {}^{13}\text{CH}_4$ minimum energy paths.^a

$s(a_0)$	-0.0050	0.0000	0.0050	$T=300\text{ K}^b$	$T=600\text{ K}^b$	$T=1500\text{ K}^b$	0.1000	0.8000
$\text{OH} + {}^{12}\text{CD}_4$								
1	3750	3748	3746	3753	3759	3768	3861	3861
2	2416	2418	2421	2431	2433	2436	2519	2802
3	2411	2415	2420	2435	2439	2443	2503	2513
4	2232	2234	2237	2251	2256	2261	2408	2513
5	1119	1123	1128	1149	1155	1162	1386	2261
6	1077	1077	1076	1076	1076	1076	1121	1458
7	1072	1071	1071	1130	1131	1132	1132	1132
8	980	979	980	993	998	1004	1064	1067
9	897	892	887	872	868	864	770	390
10	875	869	863	843	838	833	703	3
11	709	694	678	630	619	606	360	0
12	267	276	285	306	309	311	221	0
13	220	230	240	266	271	276	245	0
14	22	38	56	47	35	23	0	0
$\text{OH} + {}^{13}\text{CH}_4$								
1	3752	3748	3747	3867	3874	3885	3923	3923
2	3243	3247	3252	3272	3273	3275	3457	3795
3	3238	3243	3247	3266	3268	3270	3342	3358
4	3125	3128	3133	3165	3168	3172	3331	3358
5	1525	1520	1518	1529	1530	1533	1888	3200
6	1495	1492	1489	1485	1485	1485	1555	1663
7	1404	1408	1412	1426	1426	1427	1444	1445
8	1302	1293	1284	1252	1250	1248	1403	1445
9	1207	1201	1195	1170	1168	1166	1028	498
10	930	930	930	929	929	929	924	882
11	781	756	732	648	643	635	336	0
12	345	356	367	407	409	413	564	5
13	299	316	332	377	379	382	249	0
14	26	40	58	94	92	88	0	0

^aAll modes have a symmetry.^bColumns headed by a temperature refer to the CVT location for that temperature. The s values at these locations are given in Table V.

that the largest deviation 0.031 from the correct value corresponds to $s=0.005a_0$ for reaction (R2).

All *ab initio* calculations were carried out using the GAUSSIAN90⁴⁹ program on the Cray X-MP/EA computer at the Minnesota Supercomputer Center.

II.B. Dynamics calculations

According to TST, the rate expression is given by

$$k^{\text{TST}} = \frac{\sigma}{\beta h} \frac{Q^\ddagger}{Q^R} e^{-\beta V^\ddagger}. \quad (3)$$

In this equation, σ is the symmetry factor⁵⁰ accounting for the reaction-path degeneracy (the existence of equivalent paths), β is $(kT)^{-1}$, k is Boltzmann's constant, T is temperature, h is Planck's constant, Q^R is the reactant partition function per unit volume with the zero of energy at classical equilibrium for reactants, $V^\ddagger$ is the saddle point potential energy with the zero of energy at classical equilibrium for reactants, and $Q^\ddagger$ is the transition state partition function with the zero of energy at $V^\ddagger$ and one degree of freedom (the reaction coordinate) fixed. All symmetry numbers are omitted from rotational partition functions in

$Q^\ddagger$ and Q^R , but are included in σ (and, as discussed below, in hindered rotation partition functions). Then, the per site KIE is defined as

$$\eta = \frac{k_U/k_S}{\sigma_U/\sigma_S}, \quad (4)$$

where k_U denotes the rate coefficient for the unsubstituted reactants and k_S denotes the rate coefficient when one or more atoms are changed to one of their isotopes. The Wigner-corrected TST expression is

$$k^{\text{TST/W}} = \kappa^W k^{\text{TST}}, \quad (5)$$

where κ^W is the lowest-order semiclassical tunneling transmission coefficient due to Wigner.⁵¹

Factorization of η values determined by Eq. (5) is accomplished by

$$\eta = \eta_{\text{tun}}^{\text{TST}} \eta^{\text{TST}}, \quad (6)$$

where the tunneling contribution is

$$\eta_{\text{tun}}^{\text{TST}} = \kappa_U^W / \kappa_S^W \quad (7)$$

and the conventional TST KIE is

$$\eta^{\text{TST}} = \frac{Q_U^\ddagger/Q_U^R}{Q_S^\ddagger/Q_S^R} \quad (8)$$

Under the assumption that rotations and vibrations are separable, η^{TST} may be further factored into translational, rotational, and vibrational contributions

$$\eta^{\text{TST}} = \eta_{\text{trans}}^{\text{TST}} \eta_{\text{rot}}^{\text{TST}} \eta_{\text{vib}}^{\text{TST}} \quad (9)$$

If we now consider a generalized transition state which intersects the MEP at some value of s other than the saddle point (which is the highest-energy point on the MEP from reactants to products), we can define a generalized TST rate constant $k^{\text{GT}}(s)$ at temperature T for this dividing surface as follows:

$$k^{\text{GT}}(s) = \frac{\sigma}{\beta h} \frac{Q^{\text{GT}}(s)}{Q^R} e^{-\beta V_{\text{MEP}}(s)} \quad (10)$$

Here s is the signed distance from the saddle point of the generalized transition state along the MEP through a mass-scaled Cartesian coordinate space,^{31,41,42} $V_{\text{MEP}}(s)$ is the classical energy (also called the Born–Oppenheimer potential) along the MEP with its overall zero of energy at the reactant, and $Q^{\text{GT}}(s)$ is the partition function of the generalized transition state at s with the local zero of energy at $V_{\text{MEP}}(s)$ and again excluding symmetry numbers for rotation. In Eq. (10), all bound vibrations are quantized, but the reaction coordinate motion is treated classically.

The canonical variational transition state theory (CVT) rate coefficient k^{CVT} can be obtained from Eq. (10) by variationally minimizing the generalized transition state theory rate constant $k^{\text{GT}}(s)$ with respect to the position s of the generalized transition state along the reaction coordinate

$$k^{\text{CVT}} = \min_s k^{\text{GT}}(s), \quad (11)$$

and the semiclassical VTST rate constant to be considered in this study is given by^{26(b),31,41}

$$k^{\text{CVT/SCT}} = \kappa^{\text{SCT}} k^{\text{CVT}}, \quad (12)$$

where κ^{SCT} is a transmission coefficient that incorporates quantum effects on the reaction coordinate. This is based on a small-curvature tunneling (SCT) correction^{15,16} to the CVT prediction for the ground state of the transition state.

As mentioned above, all symmetry numbers are excluded from rotational partition functions, which are treated by classical mechanics, and we include them in σ . Then the symmetry factor for reaction (R1) is given by

$$\sigma = \frac{\sigma_{12\text{CD}_4} \sigma_{\text{OH}}}{\sigma_{\text{GT}}} \quad (13)$$

and for reaction (R2) is equal to

$$\sigma = \frac{\sigma_{13\text{CH}_4} \sigma_{\text{OH}}}{\sigma_{\text{GT}}}, \quad (14)$$

where σ_{GT} , $\sigma_{12\text{CD}_4}$, σ_{OH} , and $\sigma_{13\text{CH}_4}$ are the symmetry numbers for the generalized transition states $^{12}\text{CD}_4$, OH,

and $^{13}\text{CH}_4$, respectively. σ_{GT} and σ_{OH} are equal to 1, whereas $\sigma_{12\text{CD}_4}$ and $\sigma_{13\text{CH}_4}$ are equal to 12. A symmetry factor σ equal to 12 is obtained from Eqs. (13) and (14) for reactions (R1) and (R2), respectively. The symmetry number for reaction (R3) is already known¹⁴ to be equal to 12, so the σ_U/σ_S ratio in Eq. (4) is unity in this article.

All partition functions in Eqs. (3) and (10) are approximated here as products of translational, rotational, vibrational, and electronic partition functions. The rotational partition function required is just the product, denoted I , of the three principal moments of inertia, which can be calculated straightforwardly from the geometry.

The harmonic approximation is assumed in all cases for the calculation of the vibrational partition function, except for the internal rotational motion of the methyl group at the transition state and along the MEP where the hindered rotor^{14,15,52} model has been employed, and the vibrational information required is just the set of harmonic frequencies as a function of s . The hindered rotor partition function additionally requires σ_{ir} , the symmetry number for internal rotation, and I_r , the moment of inertia for the internal rotation^{14,15,52} (also called the reduced moment of inertia), where σ_{ir} is equal to 3 for all reactions considered here.

We assume that the electronic excitation energies and degeneracies of the generalized transition state are the same as the conventional transition state ones. In reactions (R1)–(R3), we assume no low-lying excited states of the $^2A'$ saddle point, but we correctly include the $^2\Pi_{1/2}$ excited state of OH in the reactant partition function. The excitation energy of this state of OH is 140 cm^{-1} .⁴⁰

For a deeper understanding of the per site KIE, we now apply a more detailed factorization scheme,^{6–8,53} e.g. Eq. (6) is replaced by

$$\eta = \eta_{\text{tun}} \eta_{\text{PF}} \eta_{\text{pot}}, \quad (15)$$

where

$$\eta_{\text{tun}} = \kappa_U^{\text{SCT}} / \kappa_S^{\text{SCT}}, \quad (16)$$

$$\eta_{\text{PF}} = \frac{Q_U^{\text{CVT}}/Q_U^R}{Q_S^{\text{CVT}}/Q_S^R}, \quad (17)$$

$$\eta_{\text{pot}} = \exp[-\beta(V_U^{\text{CVT}} - V_S^{\text{CVT}})]. \quad (18)$$

and PF denotes “partition function.” Significant differences emerge from the VTST treatment when it is compared to the TST one. The TST treatment neglects the potential energy contributions which cancel because the same transition state—the saddle point—applies to both isotopes, whereas the VTST calculation includes this difference. Moreover, the TST saddle point force constants are independent of isotopic substitution, while the VTST optimized transition state force constants depend on the change in structure along the MEP.

Next, η_{PF} is further factored into translational, rotational, and vibrational contributions, under the assumption that rotations and vibrations are separable, as follows:^{6,53}

$$\eta_{\text{PF}} = \eta_{\text{trans}} \eta_{\text{rot}} \eta_{\text{vib}} \quad (19)$$

There is no contribution from η_{elec} , since it is an excellent approximation to assume that electronic partition functions cancel.

Specifically,⁷

$$\eta_{\text{trans}} = Q_{\text{trans},S}^R / Q_{\text{trans},U}^R, \quad (20)$$

where $Q_{\text{trans},I}^R$ for $I=U, S$ is the translational partition function per unit volume for the reactants in the reaction under study

$$\eta_{\text{rot}} = \frac{Q_{\text{rot},U}^{\text{CVT}} / Q_{\text{rot},U}^R}{Q_{\text{rot},S}^{\text{CVT}} / Q_{\text{rot},S}^R}, \quad (21)$$

where $Q_{\text{rot},I}^{\text{CVT}}$ and $Q_{\text{rot},I}^R$ for $I=U, S$ are the generalized transition state rotational partition function and the reactant rotational partition function, respectively, and

$$\eta_{\text{vib}} = \frac{Q_{\text{vib},U}^{\text{CVT}} / Q_{\text{vib},U}^R}{Q_{\text{vib},S}^{\text{CVT}} / Q_{\text{vib},S}^R}, \quad (22)$$

where $Q_{\text{vib},I}^{\text{CVT}}$ and $Q_{\text{vib},I}^R$ for $I=U, S$ are vibrational partition functions.

Detailed analyses of the vibrational contribution of Eq. (22) are accomplished according to the scheme

$$\eta_{\text{vib}} = \eta_{\text{vib,low}} \eta_{\text{vib,mid}} \eta_{\text{vib,high}}, \quad (23)$$

where the mode classification is based on the magnitude of the vibrational frequency for the isotopically unsubstituted (U) [reaction (R3)] case. The low/mid border is taken as 100 cm^{-1} and the mid/high border is taken as 2000 cm^{-1} . Similar factorizations are performed in the analysis of the results from the conventional TST calculations.

III. RESULTS AND DISCUSSION

Both TST and VTST—through the second-order global version^{13,14} of IVTST—were applied to reactions (R1) and (R2). Second-order means that *ab initio* calculations are available not only at the reactants, saddle point, and products of the reactions under study, but also at two extra points $s=s_1$ and $s=s_2$ on each side of the saddle point and along the reaction path. As explained elsewhere,^{13,14} IVTST is based on interpolating the quantities needed for a variational transition state theory calculation and a tunneling calculation as a function of the reaction coordinate. It is not necessary to interpolate the geometry along the reaction path.

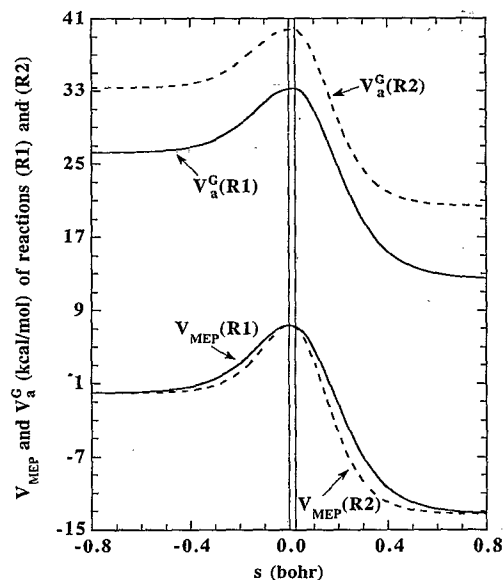


FIG. 1. Potential energy and vibrationally adiabatic ground-state potential energy of reactions (R1) and (R2) along the minimum energy path as a function of reaction coordinate. The thin vertical lines show the saddle point ($s=0$) and the location ($s=0.027a_0$) of the canonical variational transition state at 1500 K for reaction (R1).

Figure 1 shows the classical potential energy profiles $V_{\text{MEP}}(s)$ and the vibrationally adiabatic ground-state potential energy curves $V_a^G(s)$ all with overall zero of energy at the classical equilibrium energy of reactants and all as functions of the reaction coordinate s for reactions (R1) and (R2). The vibrationally adiabatic ground-state potential energy curve is expressed (for nonlinear generalized transition states) as

$$V_a^G(s) = V_{\text{MEP}}(s) + \frac{1}{2} hc \sum_{m=1}^{3N-7} \bar{\nu}_m(s), \quad (24)$$

where c is the speed of light and $\bar{\nu}_m$ is a generalized normal-mode frequency in wave numbers. The maximum of the $V_a^G(s)$ curve and the location of this maximum are denoted as V_a^{AG} and s_*^{AG} , respectively.

As mentioned above, at the MP-SAC2//MP2/adj-cc-pVTZ level, the classical forward barrier height is predicted to be 7.37 kcal/mol and the reverse barrier height is predicted to be 20.64 kcal/mol for all three reactions. In-

TABLE V. Generalized transition states' locations at various temperatures, their respective classical energy values, and their respective vibrationally adiabatic ground-state energy values for reactions $\text{OH} + {}^{12}\text{CH}_4$, $\text{OH} + {}^{12}\text{CD}_4$, and $\text{OH} + {}^{13}\text{CH}_4$.

T (K)	$\text{OH} + {}^{12}\text{CH}_4$			$\text{OH} + {}^{12}\text{CD}_4$			$\text{OH} + {}^{13}\text{CH}_4$		
	s_*^{CVT}	$V_{\text{MEP}}(s_*^{\text{CVT}})$	$V_a^G(s_*^{\text{CVT}})$	s_*^{CVT}	$V_{\text{MEP}}(s_*^{\text{CVT}})$	$V_a^G(s_*^{\text{CVT}})$	s_*^{CVT}	$V_{\text{MEP}}(s_*^{\text{CVT}})$	$V_a^G(s_*^{\text{CVT}})$
250	0.0245	7.147	39.997	0.0189	7.282	33.271	0.0220	7.185	39.899
300	0.0248	7.142	39.996	0.0197	7.274	33.265	0.0222	7.181	39.898
400	0.0253	7.133	39.993	0.0211	7.260	33.253	0.0227	7.173	39.896
600	0.0260	7.120	39.989	0.0230	7.239	33.236	0.0233	7.162	39.893
1000	0.0269	7.101	39.981	0.0251	7.215	33.218	0.0241	7.147	39.887
1500	0.0280	7.078	39.970	0.0267	7.195	33.204	0.0250	7.131	39.879
2400	0.0300	7.035	39.947	0.0284	7.172	33.189	0.0264	7.102	39.863

TABLE VI. KIEs and factors for the $\text{OH} + {}^{12}\text{CD}_4$ reaction at various temperatures.

T (K)	Method	η_{tun}	η_{trans}	η_{rot}	η_{vib}	η_{pot}	$\eta_{\text{no tun}}$	η
250	TST/W	1.60	1.18	1.82	4.67	...	10.02	16.03
	CVT/SCT	1.17	1.18	1.82	3.74	1.31	10.52	12.31
273	TST/W	1.57	1.18	1.82	3.87	...	8.32	13.05
	CVT/SCT	1.16	1.18	1.82	3.17	1.28	8.70	10.13
293	TST/W	1.54	1.18	1.82	3.37	...	7.23	11.15
	CVT/SCT	1.15	1.18	1.82	2.80	1.26	7.55	8.70
353	TST/W	1.47	1.18	1.82	2.42	...	5.19	7.62
	CVT/SCT	1.11	1.18	1.82	2.09	1.20	5.39	6.00
416	TST/W	1.40	1.18	1.82	1.88	...	4.04	5.65
	CVT/SCT	1.08	1.18	1.82	1.67	1.17	4.19	4.53
600	TST/W	1.25	1.18	1.82	1.20	...	2.59	3.25
	CVT/SCT	1.04	1.18	1.82	1.14	1.10	2.70	2.80
1000	TST/W	1.11	1.18	1.82	0.81	...	1.75	1.95
	CVT/SCT	1.01	1.18	1.82	0.81	1.06	1.85	1.87

cluding the harmonic zero-point energy contribution—calculated at the MP2/adj-cc-pVTZ level—lowers the forward barriers to 6.84 and 5.89 kcal/mol, and the reverse barriers become equal to 20.82 and 20.63 kcal/mol for reactions (R1) and (R2), respectively.

The vibrationally ground-state barrier occurs at $s_{*}^{\text{AG}} = 0.015$ and $0.020a_0$ (Fig. 1) for reactions (R1) and (R2), respectively. The adiabatic barrier height ΔV_a^{AG} is given by

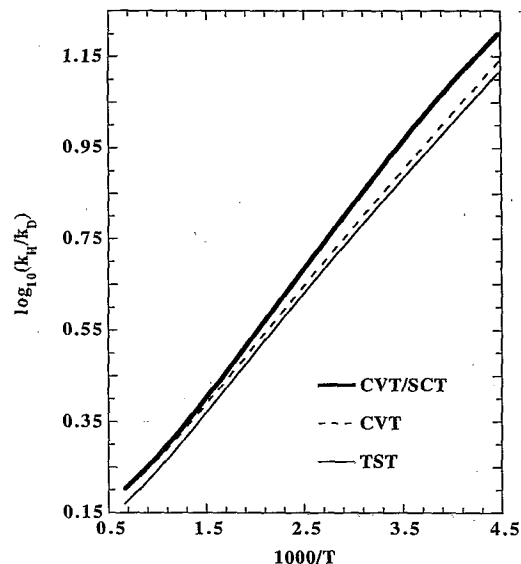
$$\Delta V_a^{\text{AG}} = V_a^{\text{AG}} - \frac{1}{2} hc \sum_{m=1}^{10} \bar{\nu}_m (s = -\infty), \quad (25)$$

where $\bar{\nu}_m$ is a reactant's vibrational frequency in wave numbers. The last term of Eq. (25) is equal to 6.94 kcal/mol for reaction (R1) and 6.00 kcal/mol for reaction (R2).

For interpretative reasons, Table V lists the location of the generalized transition states s_{*}^{CVT} for seven different temperatures, namely, 250, 300, 400, 600, 1000, 1500, and 2400 K, their classical energy values $V_{\text{MEP}}(s_{*}^{\text{CVT}})$, and their vibrationally adiabatic ground-state energy values $V_a^G(s_{*}^{\text{CVT}})$, respectively, for reactions (R1)–(R3). For each of the above temperatures, reaction (R1) always has the largest $V_{\text{MEP}}(s_{*}^{\text{CVT}})$ value and reaction (R3) the smallest

TABLE VII. KIEs and factors for the $\text{OH} + {}^{13}\text{CH}_4$ reaction at various temperatures.

T (K)	Method	η_{tun}	η_{trans}	η_{rot}	η_{vib}	η_{pot}	$\eta_{\text{no tun}}$	η
250	TST/W	1.000	1.047	0.971	0.988	...	1.005	1.005
	CVT/SCT	1.031	1.047	0.971	0.889	1.079	0.976	1.005
273	TST/W	1.000	1.047	0.971	0.988	...	1.005	1.005
	CVT/SCT	1.027	1.047	0.971	0.897	1.074	0.979	1.005
293	TST/W	1.000	1.047	0.971	0.988	...	1.005	1.005
	CVT/SCT	1.024	1.047	0.971	0.903	1.069	0.981	1.005
353	TST/W	1.000	1.047	0.971	0.988	...	1.005	1.005
	CVT/SCT	1.017	1.047	0.971	0.918	1.059	0.988	1.005
600	TST/W	1.000	1.047	0.971	0.987	...	1.004	1.004
	CVT/SCT	1.004	1.047	0.971	0.949	1.036	1.000	1.004
1000	TST/W	1.000	1.047	0.971	0.986	...	1.002	1.002
	CVT/SCT	1.001	1.047	0.971	0.962	1.024	1.001	1.002

FIG. 2. Logarithm of the deuterium KIE for reactions (R1) and (R3) vs $1000/T$.

one, whereas reaction (R1) always has the smallest $V_a^G(s_{*}^{\text{CVT}})$ value and reaction (R3) the largest one.

Factorizations of the deuterium KIE at six different temperatures in terms of translational, rotational, and vibrational motions, tunneling corrections, and the classical potential energy along the reaction path are given in Table VI. As mentioned above, there is a potential energy contribution to the KIEs only in the CVT calculations. This comes from the different classical energies of the generalized transition states for the substituted and unsubstituted reactions. Similar factorizations of the ${}^{13}\text{C}$ KIE are listed in Table VII. Figures 2 and 3 compare different levels of dynamical theory for the two kinetic isotope effects.

In order to test the sensitivity of the $k_{\text{H}}/k_{\text{D}}$ calculations to the theoretical assumptions, we repeated these calculations at the MP2/adj-cc-pVTZ level, i.e., without the SAC correction. The only difference in these calculations is that $V_{\text{MEP}}(s)$ is 0.59 kcal/mol higher at the saddle point and the two nearby points when the scaling is not included. Table VIII shows the resulting KIEs. The difference is less than 1% for $T \geq 400$ K, increasing to 5% at 300 K and

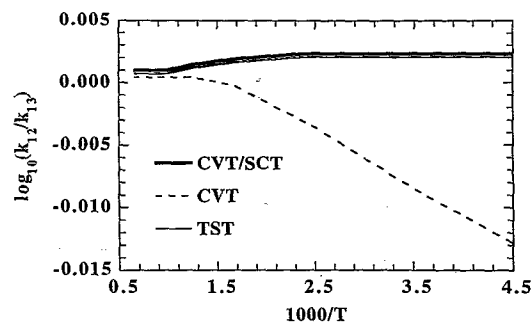
FIG. 3. Logarithm of the ${}^{13}\text{C}$ KIE for reactions (R2) and (R3) vs $1000/T$.

TABLE VIII. Final calculated and experimental KIEs for reaction (R1) at various temperatures.

Temperature (K)	MP-SAC2	MP2
223	15.9	18.7
250	12.3	13.7
293	8.70	9.18
300	8.27	8.68
400	4.82	4.85
416	4.53	4.54
600	2.80	2.80
1000	1.88	1.88
1500	1.59	1.59
2400	1.45	1.45

18% at 223 K. Thus the results are not overly sensitive to this scaling step. In the rest of the paper, we only consider our final results, which include the scaling.

III.A. Tunneling contribution

Because of the difference in the imaginary frequency at the saddle point upon substitution of $^{12}\text{CH}_4$ with $^{12}\text{CD}_4$, the Wigner method predicts an $\eta_{\text{tun}}^{\text{TST/W}}$ factor of 1.60 for reaction (R1) at 250 K. The SCT contribution for the same reaction is smaller, predicting an $\eta_{\text{tun}}^{\text{CVT/SCT}}$ factor equal to 1.17 at 250 K.

For reaction (R2), however, the $\eta_{\text{tun}}^{\text{TST/W}}$ factor equals 1.00 at any temperature, e.g., $\eta_{\text{tun}}^{\text{TST/W}}$ is equal to 1.0002 at 250 K, since the value of the imaginary frequency is practically the same for the $^{13}\text{CH}_4\text{OH}$ and $^{12}\text{CH}_4\text{OH}$ saddle points [in particular it is 1829.7 cm^{-1} for reaction (R2) and 1829.9 cm^{-1} for reaction (R3)], whereas the $\eta_{\text{tun}}^{\text{CVT/SCT}}$ factor is equal to 1.03 at 250 K, and gradually approaches the minimum value of 1.00 as the temperature rises above 1000 K. Since one-dimensional tunneling methods, such as the Wigner method, often prove to be qualitatively incorrect,^{7,8} caution should be used in their application and interpretation.

III.B. Translational contribution

The translational motion contribution is equal to the $3/2$ power of the ratio of the reduced masses of the relative translational motion of reactants in the substituted and unsubstituted reactions, and both TST and CVT predict the same small normal contribution of 1.18 for reaction (R1) and of 1.05 for reaction (R2).

III.C. Rotational contribution

The rotational motion contribution to the deuterium KIEs results from two square root ratios of products of principal moments of inertia, namely, one that depends on the $^{12}\text{CD}_4$ and $^{12}\text{CH}_4$ structures and a second one that depends on the $^{12}\text{CD}_4\text{OH}$ and $^{12}\text{CH}_4\text{OH}$ generalized transition state structures. However, in conventional TST, the generalized transition state structure is replaced by the saddle point one. The OH radical does not contribute to the $\eta_{\text{rot}}^{\text{TST}}$ or $\eta_{\text{rot}}^{\text{CVT}}$ factors, since there is no isotopic substitution

in the OH species. The calculations indicate that for the deuterium KIEs, both $\eta_{\text{rot}}^{\text{TST}}$ and $\eta_{\text{rot}}^{\text{CVT}}$ are equal to 1.82 for a wide temperature range.

The rotational contribution to the ^{13}C KIEs comes from similar factors as for the deuterium one just discussed, except involving the $^{13}\text{CH}_4$ and $^{13}\text{CH}_4\text{OH}$ structures instead of the $^{12}\text{CD}_4$ and $^{12}\text{CD}_4\text{OH}$ ones. For the ^{13}C KIEs, both $\eta_{\text{rot}}^{\text{TST}}$ and $\eta_{\text{rot}}^{\text{CVT}}$ give an inverse contribution and are equal to 0.97 in the temperature range from 250 to 1000 K.

III.D. Vibrational contribution

Large differences between $\eta_{\text{vib}}^{\text{TST}}$ and $\eta_{\text{vib}}^{\text{CVT}}$ are found at low temperatures for reaction (R2), which indicate a considerable variational effect on the vibrational contribution to the deuterium KIE. However, both theories predict a normal vibrational contribution according to Table VI in the temperature range from 250 to 600 K, and as we move from atmospheric to medium temperatures, the conventional vibrational contribution becomes closer to the variational one.

To better understand the $\eta_{\text{vib}}^{\text{TST}}$ and $\eta_{\text{vib}}^{\text{CVT}}$ factors, we factor them according to Eq. (23), where $\eta_{\text{vib,low}}$ contains contributions from modes with frequencies below 100 cm^{-1} (mode 14), $\eta_{\text{vib,mid}}$ contains contributions from modes with frequencies within the range $100\text{--}2000\text{ cm}^{-1}$ (modes 5–13), and $\eta_{\text{vib,high}}$ contains contributions from modes with frequencies greater than 2000 cm^{-1} (modes 1–4). These factors are given in Table IX for both levels of theory.

The $\eta_{\text{vib,low}}$ and $\eta_{\text{vib,mid}}$ factors in Table IX lead to inverse contributions to the deuterium KIE with $\eta_{\text{vib,mid}}$ being more significant than $\eta_{\text{vib,low}}$. The lowest-frequency mode of the transition states, which is the one responsible for the $\eta_{\text{vib,low}}$ factor and which corresponds to the internal rotational motion of the methyl group at the transition state and along the MEP, has been treated according to the hindered rotor^{14,15,52} model. Table IX clearly shows the importance of the variational effect on the vibrational contribution to the deuterium KIEs. There is a significant difference between $\eta_{\text{vib,high}}^{\text{TST}}$ and $\eta_{\text{vib,high}}^{\text{CVT}}$ (almost a factor of 2 at 250 K, decreasing to 12% at 1000 K) and another one (14%–36%) between $\eta_{\text{vib,mid}}^{\text{TST}}$ and $\eta_{\text{vib,mid}}^{\text{CVT}}$; these differences illustrate that there are much larger variational effects resulting from particular groups of vibrational frequencies than the overall variational effect arising from all the vibrational contributions taken together. The partial cancellation of individual mode differences results in a small net difference between the η^{TST} and η^{CVT} KIEs, as seen in the last column and previously⁸ in another case.

Variational effects on the vibrational contributions to the ^{13}C KIEs do not result in large differences between $\eta_{\text{vib}}^{\text{TST}}$ and $\eta_{\text{vib}}^{\text{CVT}}$, even at low temperatures. In the temperature range from 250 to 1000 K, both TST and CVT calculations predict an inverse vibrational contribution according to Table VII. As we move from the atmospheric temperature range to the medium temperature region, the conventional theory's vibrational contribution is getting closer to the variational one.

TABLE IX. Factorization of η_{vib} for the $\text{OH} + {}^{12}\text{CD}_4$ and $\text{OH} + {}^{13}\text{CH}_4$ reactions at various temperatures.

T (K)	Method	$\eta_{\text{vib,low}}$	$\eta_{\text{vib,mid}}$	$\eta_{\text{vib,high}}$	η_{vib}
$\text{OH} + {}^{12}\text{CD}_4$					
250	TST	0.87	0.49	11.00	4.67
	CVT	0.88	0.67	6.40	3.74
273	TST	0.87	0.50	8.99	3.87
	CVT	0.88	0.66	5.47	3.17
293	TST	0.87	0.50	7.74	3.37
	CVT	0.88	0.66	4.87	2.80
353	TST	0.87	0.51	5.47	2.42
	CVT	0.88	0.64	3.72	2.09
416	TST	0.87	0.51	4.23	1.88
	CVT	0.88	0.62	3.06	1.67
600	TST	0.87	0.51	2.73	1.20
	CVT	0.88	0.59	2.18	1.14
1000	TST	0.87	0.50	1.87	0.81
	CVT	0.88	0.57	1.64	0.81
$\text{OH} + {}^{13}\text{CH}_4$					
250	TST	0.998	0.968	1.023	0.988
	CVT	1.001	0.970	0.916	0.889
273	TST	0.998	0.970	1.021	0.988
	CVT	1.001	0.972	0.922	0.897
293	TST	0.998	0.972	1.019	0.988
	CVT	1.001	0.974	0.926	0.903
353	TST	0.998	0.975	1.016	0.988
	CVT	1.001	0.977	0.938	0.918
600	TST	0.998	0.979	1.010	0.987
	CVT	1.001	0.985	0.962	0.949
1000	TST	0.998	0.980	1.008	0.986
	CVT	1.002	0.989	0.971	0.962

For interpretative purposes, we further factor both $\eta_{\text{vib}}^{\text{TST}}$ and $\eta_{\text{vib}}^{\text{CVT}}$ for the ${}^{13}\text{C}$ KIE according to Eq. (23), while following the same mode classification scheme as above. Both TST and CVT calculated factors are listed in Table IX, and they differ from each other most for the high vibrational frequencies.

The lowest-frequency mode of the transition states—the one associated with the internal rotational motion of the methyl group at the saddle point and along the MEP, and which has been treated according to the hindered rotor model—provides a negligible contribution in both the variational and conventional calculations of the ${}^{13}\text{C}$ KIEs. The middle-frequency modes are responsible for a slight inverse contribution, about 3% at low T , but only 1%–2% at high T . The TST high-frequency mode's contribution is normal but only 1%–2%, whereas the CVT high-frequency mode's contribution is inverse and is as large as 8% at low T , although it decreases to 3% at 1000 K. Thus the only noticeable variational effect from a particular group of frequencies in this case is due to the high-frequency modes.

III.E. KIEs on Arrhenius parameters

Table X shows the effect of deuterium substitution on the Arrhenius parameters for an atmospheric (250–300 K) and a medium (600–1000 K) temperature region. $A_{\text{H}}/A_{\text{D}}$ values significantly less than 0.5–0.7 are often considered to be an indication of tunneling.^{5(b)} Although the present reactions are dominated by tunneling [e.g.,

TABLE X. Kinetic isotope effects on Arrhenius parameters for the $\text{OH} + {}^{12}\text{CD}_4$ and $\text{OH} + {}^{13}\text{CH}_4$ reactions in two temperature regions.

T (K)	Method	$\text{OH} + {}^{12}\text{CD}_4$		$\text{OH} + {}^{13}\text{CH}_4$	
		$A_{\text{H}}/A_{\text{D}}$	$E_{\text{a,D}} - E_{\text{a,H}}$ (kcal/mol)	$A_{\text{H}}/A_{\text{D}}$	$E_{\text{a,D}} - E_{\text{a,H}}$ (kcal/mol)
250–300	TST	1.08	1.11	1.006	−0.001
	TST/W	1.34	1.23	1.006	−0.001
	CVT	1.09	1.13	1.013	−0.019
	CVT/SCT	1.14	1.18	1.001	0.002
600–1000	TST	0.97	1.17	0.999	0.006
	TST/W	0.90	1.52	0.999	0.006
	CVT	1.05	1.12	1.003	−0.004
	CVT/SCT	1.03	1.19	1.000	0.005

$\kappa^{\text{SCT}} = 26.9, 8.7$, and 1.6 at $T = 250, 300$, and 600 K, respectively, for reaction (R3)¹⁴], we find $A_{\text{H}}/A_{\text{D}} \geq 0.9$ at all levels of theory and $A_{\text{H}}/A_{\text{D}} > 1$ for our most accurate level of theory. This illustrates the danger of an Ockham's razor approach to tunneling, i.e., tunneling may be very important even when the "telltale signs" are missing. The final CVT/SCT calculations predict $A_{\text{H}}/A_{\text{D}}$ values that differ from the conventional TST ones by 6% and $E_{\text{a,D}} - E_{\text{a,H}}$ values that differ from the conventional TST ones by 0.002–0.07 kcal/mol. However, the differences between TST and CVT/SCT calculations are larger for individual reactions [e.g., at 600 K, the CVT/SCT rate differs from the conventional TST one by 19% and 25% for reactions (R1) and (R3), respectively]; thus we see that a large part of the error cancels in the KIE.

The ${}^{13}\text{C}$ isotopic substitution effect on the Arrhenius parameters is always negligible for both temperature regions (Table X) since the KIE value is very close to unity. However, the CVT inverse isotope effect is reflected in the activation energy as a slightly negative contribution.

III.F. Comparison to available experimental data

There is only one published experimental value⁹ for the deuterium KIE in reaction (R1). This experimental value is 11 at 416 K, which is a factor of 2.4 higher than the value calculated here (Table XI). For the ${}^{13}\text{C}$ isotope effect in reaction (R2), the calculated values are in excellent agreement with all experimental¹² ones, which are available at three different atmospheric temperatures, as listed in Table XI.

(Addendum: After this paper was submitted for publication, we sent a preprint to Tully and he sent us new unpublished experimental results. Figure 4 shows a comparison of the present predictions for $k_{\text{H}}/k_{\text{D}}$ to an Arrhenius fit to the new experimental results. The agreement is much better than with the previous experiment. This comparison will be discussed further in a paper⁵⁴ now in preparation by the experimentalists.)

IV. CONCLUSIONS

In this article, we showed that variational transition state theory calculations with multidimensional tunneling probabilities may be carried out for kinetic isotope effects

TABLE XI. Final calculated and experimental KIEs for reactions (R1) and (R2) at various temperatures.

Temperature (K)	OH + $^{12}\text{CD}_4$		OH + $^{13}\text{CH}_4$	
	Theory ^a	Expt. ^b	Theory ^a	Expt. ^c
223	15.9	n/a ^d	1.005	n/a
273	10.1	n/a	1.005	1.005
293	8.70	n/a	1.005	1.005
298	8.39	n/a	1.005	n/a
300	8.27	n/a	1.005	n/a
353	6.00	n/a	1.005	1.005
400	4.82	n/a	1.005	n/a
416	4.53	11	1.005	n/a
800	2.16	n/a	1.003	n/a
1500	1.59	n/a	1.002	n/a
2400	1.45	n/a	1.001	n/a

^aThis study.^bReference 9.^cReference 12.^dn/a—not available.

(KIEs) on the basis of very large scale electronic structure calculations for reactants, products, and three points along the reaction path for each isotopic version of the reaction. We also showed that deuterium KIEs for reaction (R1) and ^{13}C KIEs for reaction (R2) calculated by interpolated variational transition state theory with small-curvature tunneling coefficients are significantly different from KIEs calculated for the same reactions by conventional TST. Moreover, our calculations show that when generalized transition states are defined and their optimum location is determined variationally, the location is often strongly dependent on isotopic substitution. The calculated k_{12}/k_{13} KIEs agree excellently with the available experimental values,¹² whereas the $k_{\text{H}}/k_{\text{D}}$ KIEs are more than a factor of 2 lower than the sole available experimental value.⁹ More experimental tests of the $k_{\text{H}}/k_{\text{D}}$ KIEs will be most welcome. (*Addendum:* See last paragraph of Sec. III.F.)

Our detailed analysis of the contributions of the individual groups of vibrational frequencies show that significant (33% to factor of 11) contributions to the deuterium KIE arise from both mid- and high-frequency modes, whereas vibrational contributions to the ^{13}C KIE are always within 11% of unity. For both KIEs, these major

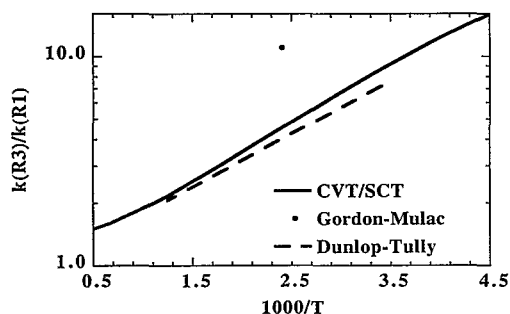


FIG. 4. Deuterium KIE for reactions (R1) and (R3) vs $1000/T$. The present results are at the CVT/SCT level and the other results are from experiments.

contributions are significantly affected (e.g., factors of 1.4, 1.7, and 1.12 at 250 K) by variational optimization of the transition state location.

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